# Phases of the Moon

LAB LESSON

## OBJECTIVES

In this lesson, students will develop their understanding of the phases of the Moon. Constructing their own moon wheel using the information presented in the ClickView video will help them visualise why this phenomenon occurs.

## ACARA CONTENT DESCRIPTION

**Predictable phenomena on Earth, including seasons and eclipses, are caused by the relative positions of the sun, Earth and the moon (ACSSU115)**

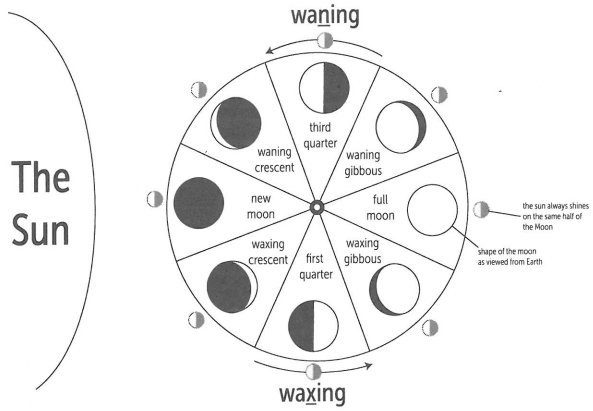
- investigating natural phenomenon such as the phases of the Moon
- modelling the relative movements of the Earth, sun and moon and how natural phenomenon such as phases of the Moon occur

## LESSON PLAN

| Activities | Resources |
|---|---|
| **Activity 1: Lightning Writing**<br>Lightning writing is a strategy used to encourage students to write as much as they can about a topic in a set period of time.<br>Have students respond to the questions on the first slide of the presentation using lightning writing. Discuss answers as a class using slides 2-5 of the presentation.<br>10 | • Presentation: Phases of the Moon |
| **Activity 2: It's Just a Phase**<br>Give out the It's Just a Phase 1 and 2 worksheets to students. Play the video and allow students time to complete the shading and labelling of the Moon Wheel (page 40). Students should only start to fill in their worksheet from 1:45 of the video. Pause the video if required to allow for catch up. Then, cut out the wheel on the It's Just a Phase 1 worksheet and secure it with split pins.<br>20 | • Presentation: Phases of the Moon<br>• ClickView video *The Phases of the Moon* **https://clickv.ie./w/lp/10**<br>• Photocopies of the It's Just a Phase 1 and 2 worksheets (on separate pages)<br>• Split pins and scissors |
| **Activity 3: Modelling the Moon's Phases**<br>Carry out the activity with the following instructions:<br>1. Pierce the polystyrene ball with the pencil so it appears lollipop shaped. This is Earth's Moon.<br>2. Turn the light source on and place it in the middle of the room. This is the Sun.<br>3. Turn off the lights in the classroom.<br>4. Select a volunteer to stand approximately 2 m from the light source and hold the Moon with their arm extended, such that they can still see the light. The student's view of the Moon represents that as seen from Earth. This student will start the activity facing the light source. The Moon should be in its new moon phase.<br>5. Discuss observations as a class.<br>6. Have the student holding the Moon move slowly in a counter clockwise direction until the next moon phase is identifiable. Continue this until all the moon phases have been achieved. Students should be checking each phase with their moon wheels.<br>7. Once back at the new moon starting position, have another student repeat the activity, but this time holding a digital camera in front of them to film the moon phases.<br>15 | • Polystyrene ball (approx. 8 cm in diameter)<br>• Pencil<br>• Light source with a bright, clear bulb (lamp or overhead projector)<br>• Digital camera |
| **Activity 4: Moon Bingo Quiz**<br>Print the Moon Bingo Cards and give a piece to each student pair. Ask students to, at random, assign each of the 24 words and phrases shown on the last slide of the presentation to the blank boxes on the worksheet. Then using the Moon Bingo Quiz on the right, ask students to pick a number between 1-24. Ask the class to guess the missing word(s) for each numbered statement. The underlined words represent the answers. The first student to get a vertical or horizontal line of 5 boxes crossed out calls "Bingo!".<br>15 | • Presentation: Phases of the Moon<br>• Photocopies of the Moon Bingo Cards<br>• Moon Bingo Quiz |

## ANSWERS

It's Just a Phase




## EXTRAS

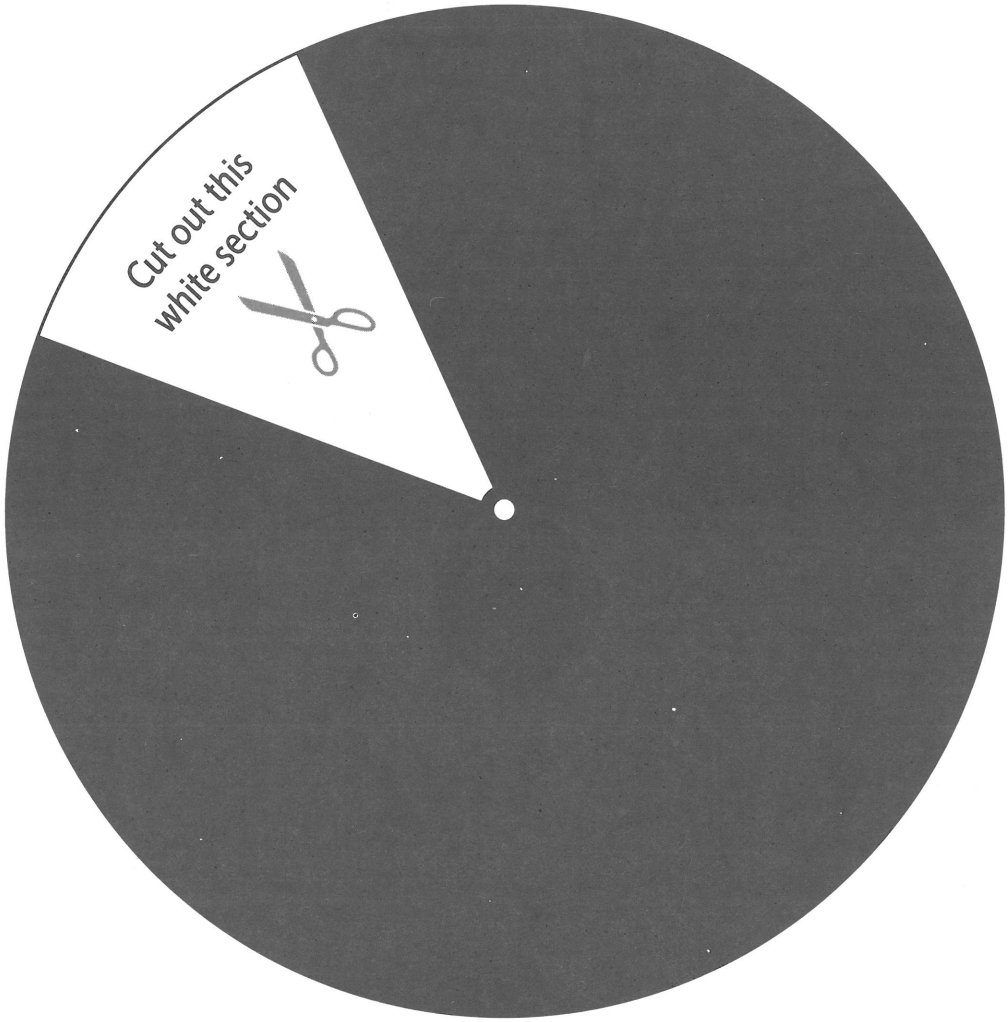
Moon Bingo Quiz

| | What's the Missing Word(s)? |
|---|---|
| 1 | The phase after the *first quarter* is known as the waxing gibbous. |
| 2 | The start of the moon cycle is known as the *new* moon. |
| 3 | When the visible part of the Moon is getting bigger, it is also known as *waxing*. |
| 4 | The *Moon* orbits around Earth. |
| 5 | *Earth* orbits around the Sun. |
| 6 | We are only able to see one *side* of the Moon. |
| 7 | The Moon orbits around Earth in around *27.3* days. |
| 8 | When the visible part of the Moon is getting smaller, it is known as *waning*. |
| 9 | The *visible* part of the Moon is determined by the location of the Moon in relation to the Sun. |
| 10 | A *gibbous* moon is when the visible part of the Moon is greater than a half moon. |
| 11 | A *crescent* moon is when the visible part of the Moon is less than a quarter moon. |
| 12 | A *full moon* is the phase when the visible part of the Moon is at its largest. |
| 13 | The Moon *reflects* light from the Sun. |
| 14 | The Moon is *smaller* than Earth. |
| 15 | This phase before the waning crescent is known as the *third quarter*. |
| 16 | There are *eight* distinct moon phases. |
| 17 | The Earth has *one* natural satellite. |
| 18 | The *Sun* is the reason we can see the Moon. |
| 19 | The Moon orbits *counter clockwise* around Earth. |
| 20 | The *phases* of the Moon are a result of the location of the Moon with respect to the Sun. |
| 21 | The Earth is *384 400* km away from the Moon. |
| 22 | The *shape* of the Moon tells us which phase the Moon is in. |
| 23 | The Moon travels around Earth in a circle, known as an *orbit*. |
| 24 | The Moon is also known as a *satellite*, which orbits around Earth. |



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# It's Just a Phase 1




## Instructions:

You will need the It's Just a Phase 2 worksheet to complete this task.

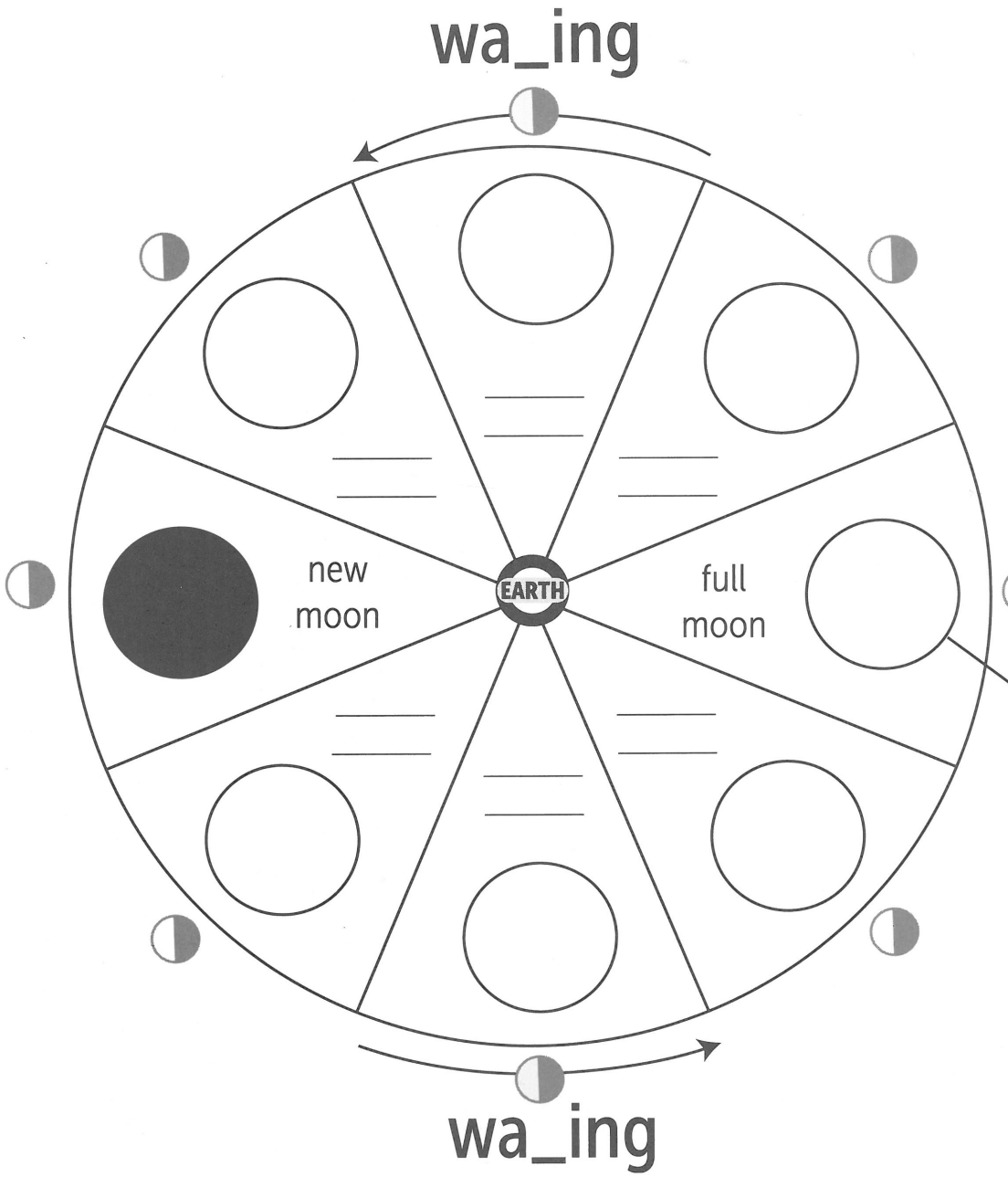
1. Cut out the black circle on this page. Be sure to also cut out the white wedge.
2. Complete the missing letters on the It's Just a Phase 2 worksheet.
3. As you watch the video, shade the moon for each of the moon phases as how it would appear when viewed from Earth.
4. Using a split pin, connect your black circle to the centre of the moon wheel.





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# The Sun






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# Moon Bingo Cards

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